

Market Benchmark — Reception Robot

Design references and proposed inspiration for the Reception & Luggage Service Robot

Objective

Study existing indoor service and mobile robots to identify useful mechanical and interaction concepts. The final robot will remain an original design, adapted to the project's dimensions, components, manufacturing constraints and future ROS2/Gazebo simulation.

Our main constraints

Parameter	Target
Overall height	≈ 1500 mm
Main base	750 × 400 mm
Base height	≈ 400–500 mm
Drive wheels	2 × Ø200 mm (8 inch)
Caster wheels	2
Payload	20–30 kg
Battery	260 × 160 × 135 mm
LiDAR	RPLIDAR A1M8
Camera	CSI IMX519, above LCD
Display	10.1-inch Waveshare HDMI LCD

Reference robots

Robot	What to inspire from	Priority
ROMR	Chassis, modular structure, aluminum profiles, standard parts, maintenance and CAD/URDF workflow	★★★★★
TurtleBot 4	RPLIDAR A1M8, mobile base, sensor organization, ROS2/URDF and simulation	★★★★★
KettyBot Pro	Reception/service design, LCD, camera, interaction and luggage/serving area	★★★★■
PUDU T300	Industrial base, stability, payload transport, wheels and maintenance	★★★★■
SMARTmBOT	Simple modular construction, 3D-printed parts and open-source educational approach	★★★██

Proposed architecture

Lower mobile base → battery and electronics low in the chassis → RPLIDAR A1M8 → luggage platform for 20–30 kg → removable central mast → 10.1-inch LCD → IMX519 camera above the display. The design should use mostly flat panels, aluminum profiles, standard fasteners and simple brackets.

Subsystem decisions

Subsystem	Reference	Our direction
Chassis	ROMR + T300	Simple rectangular modular chassis with easy access.
Wheels	ROMR + TurtleBot 4	2 powered Ø200 mm wheels + 2 casters.
Battery/electronics	ROMR	Heavy components low; removable panels and cable routing.
Luggage	KettyBot Pro + T300	Large anti-slip platform with raised edges.
Mast	Original	Simple removable rectangular/tubular mast.
LCD + camera	KettyBot Pro	Front LCD, IMX519 directly above.

Subsystem	Reference	Our direction
LiDAR	TurtleBot 4	RPLIDAR A1M8 with unobstructed scan field.
ROS2/URDF	TurtleBot 4 + ROMR	Separate CAD modules/links for URDF/Xacro.

Proposed workflow

Market benchmark → 2–3 concepts → review/selection with Amine → master layout → detailed CAD → assembly validation → drawings → STEP/STL → URDF/Xacro → RViz/Gazebo.

Sources

ROMR — <https://arxiv.org/abs/2210.01627>

ROMR full article — <https://pmc.ncbi.nlm.nih.gov/articles/PMC10197097/>

TurtleBot 4 — <https://turtlebot.github.io/turtlebot4-user-manual/>

PUDU T300 — <https://www.pudurobotics.com/en/products/pudut300>

KettyBot Pro — <https://academy.pudutech.com/en/docs/chan-pin-jian-jie-hu-lu-pro>

SMARTmBOT — <https://arxiv.org/abs/2203.08903>